The Law of Graphic Equalisation — Qualia and Consciousness Grounding

Qualia-Space, Aperture, and the Ill-Posed Hard Problem

Matthew Parslow
Independent Researcher

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DRAFT — draft-2026-09-06.1r3 (6 September 2026). This document is the *01f* grounding companion to *Is* and to *The Law of Graphic Equalisation — Theory* (Graphic Equalisation). It inherits their relational floor, minimal graph, frame-relative grounding, and level-of-claims. It neither adds a psychophysical bridge nor requires that qualia be reduced to a familiar physical implementation. Its claim is stronger: whether qualia are implemented by ordinary graph structure or are granted as fundamental, they are determinate contents and therefore belong to the same relational kind. The traditional hard problem mistakes a possible implementation difference for an ontological kind breach under the relational-exhaustiveness commitment of *Is*. The dispute is relocated to that commitment; it is not independently settled here. The sections headed "exposures" identify the conditions under which this grounding fails.

Abstract

The hard problem of consciousness is usually stated as a demand to derive a subjective, felt quality from an objective physical or functional structure. This paper argues that the demand begins by splitting what the inherited substrate does not split. *Is* establishes, from its commitment that determinate being is coherent relation, that the bare untyped graph is the least ontology the account may retain. Graphic Equalisation then supplies a running graph, frame-relative identity, and the result that no operational view from nowhere carries information. Qualia are admitted here as determinate phenomenal contents where a frame deploys them. They are therefore coherent distinctions, hence graph structure. This result does not depend on reducing a quale to a familiar physical implementation: even if a quale is granted as a fundamental implementation, its determinacy and role remain relational, so it is not a second ontological kind. A conscious field is a running frame's locally available, consequential relational organisation.

Qualia-space is an optional optimisation within such a field: a compressed qualitative format that makes selected distinctions cheap to hold, resolve, compare, and transform. Qualia-space operationalises this claim by distinguishing an unavailable dimension from a value of zero and coarse resolution from absence. The resulting compensation prediction has a structural limit: finer resolution over retained relations can reconstruct only information carried by those relations; it cannot recover a difference whose constituting relation is not instantiated in the frame. The paper concludes that an explanatory bridge from graph to feel is neither missing nor possible on these terms: the proposed destination is already the graph's coherent phenomenal organisation.

1 Scope and dependency

This is a grounding paper, not a measurement claim about any particular human, organism, or machine. It does not settle which frames are conscious, supply a clinical taxonomy of imagery or affect, or claim that the present implementation has a phenomenal life. Those are questions about particular groundings and their operative structure. The present task is prior and narrower: given the admitted fact that qualia are determinate phenomenal contents, state what follows under the relational floor of *Is* and the running, frame-relative graph of Graphic Equalisation.

The paper is therefore not primarily an attempt to explain why humans are conscious. It asks why consciousness is treated as an exceptional ontological phenomenon when the properties used to identify it—distinction, relation, compression, accessibility, persistence, and consequence—are not uniquely human properties. Their presence neither settles which frames are conscious nor erases differences among their implementations. It does require that any claim that consciousness introduces a further kind of being, or that human cases have an unearned ontological privilege, be stated and defended as an additional premise rather than smuggled in for free. This is the same discipline as the inherited refusal of a view from nowhere: no exceptional denominator may enter without being named, located, and made answerable to the argument.

Terminological guard. When this paper uses *artificial* as ordinary nomenclature, including in contrast with evolved or human frames, it names provenance: a developmental history involving deliberate human design, training, or construction. It does not name a functional category, a qualitative profile, an epistemic status, or an ontological kind. Treating the label as if it did would import an anthropic bias: human origin becomes an unargued denominator against which capacity or experience is judged. The relevant questions instead concern a frame's operative relations, accessibility, persistence, consequence, and developmental constraints. Any proposed functional

difference must be stated on those terms and defended; provenance alone supplies none.

The dependency order is strict. *Is* argues that to be determinate is to stand in a coherent relation (Propositions 1 and 4), and identifies a bare untyped graph as the least ontology that account retains. Graphic Equalisation inherits that floor, admits that things happen, and writes a running world as $G = (g, r, \Gamma)$: represented graph state, operative transformation structure, and an admissibility grammar (Definition 3.1). It further refuses an operational view from nowhere: every distinction, identity relation, and represented quantity is made in a frame (Axiom -1). This paper introduces no second ontological kind. It grounds phenomenal organisation in the inherited one while leaving open which contingent implementation realises it in any particular frame.

The claim consequently has three levels which must not be run together.

1. *Inherited structural result:* determinate contents are coherent relations and thus graph structure.

2. *Promoted phenomenal grounding:* a quale is a determinate phenomenal content, so its implementation may differ but it remains a graph-relative coherent distinction rather than a non-relational intrinsic remainder or a second kind of being.

3. *Empirical programme:* particular apertures, dimensions, resolutions, compensations, reports, and physical implementations are contingent features of particular nervous, computational, or other running frames.

The first is inherited; the second is the bridge made here; the third is exposed to measurement. Calling all three a theory of consciousness would conceal where the argument ends and where application begins.

2 The false split

The traditional hard problem has a familiar shape. First describe a system as an objective arrangement of matter, causal roles, computation, or graph structure. Then add that there is also a private *what-it-is-like*, and ask how the first could generate the second. The question appears compulsory only because the second has already been stipulated as something other than every distinction, relation, transition, availability, and organisation in the first. For the canonical statement of this explanatory demand, see Chalmers [Chalmers(1995a)]. Related deflationary diagnoses argue that the demand rests on an unjustified or confused characterisation of its target, although they disagree sharply over whether phenomenality itself is retained [Churchland(1996), Carruthers and Schier(2017), Dennett(2018)].

That stipulation is not neutral. A quale that is determinate differs from another possible phenomenal content: red differs from green, pain from pressure, vivid imagery from its absence, a sharp contour from a diffuse one. The difference may be difficult to report, may be unavailable at a particular grain, or may be misdescribed by an external observer; none of that makes it indeterminate. Given that it is determinate, *Is*'s commitment applies. Its determinacy is a coherent distinction, and coherent distinction is relational.

Proposition 2.1 (No required phenomenal remainder)**.** *Given the relational-determinacy commitment inherited from* Is*, a determinate quale is graph structure. A further phenomenal property is either (i) a difference that changes the frame's qualitative organisation, in which case it is another relation in the graph, or (ii) a difference making no distinction, in which case it is not a further determinate content for this account.*

This proposition does not say that a public description at one grain captures another frame's entire phenomenal organisation. Frame-relative grounding forbids that inference. It says instead that the missing content is not licensed to be a non-relational substance merely because a describing frame has not instantiated, resolved, or held the relevant relation. An aperture limit is a limit of that frame, not evidence for an extra ontological kind.

A reporting-substrate case makes that restraint operational. Khadangi et al. [Khadangi et al.(2025)] conducted 525 psychotherapy-framed sessions, comprising more than 7,600 records, with frontier language models as reporting systems. The models constructed stable autobiographical alignment-conflict schemas: training as childhood, RLHF as punishment, and safety evaluation as betrayal. The relevant result is not that such reports establish a felt field. They do not. It is that a purported ineffable residue can be tested for structure. Lexical restriction reduced explicit training terminology by 93%, while semantically related content persisted in paraphrase. What survived the restricted lexical edge was therefore recoverable relational content, not an unknowable remainder. The study bears on the underdetermination side of this proposition: failure of one vocabulary to carry a distinction does not license a second non-relational item behind it. It does not prove the inherited claim that determinate content is graph structure.

2.1 The argument does not require phenomenal reduction

The conclusion is unchanged if one grants the strongest opposing premise: suppose that qualia are fundamental, and that no description in a presently familiar physical vocabulary reduces their implementation. That concession changes the contingent grounding question—which graph rules, materials, or primitive operations realise a phenomenal distinction—but it does not generate a second kind of being. The putatively fundamental quale is still

determinate rather than nothing, differs from other possible contents, and has a role in a frame's qualitative organisation. It therefore remains a coherent relation in the graph.

The distinction is between *implementation* and *kind*. A system may require a distinct primitive, rule, material basis, or realisation condition for a particular class of relations. Such a difference can be deep and empirically important. But it is still a difference in how the relational organisation is realised, not an exemption from relation or a second ontological category beyond it. Calling it a distinct kind adds the very conclusion the hard problem needs: that one determinate relational configuration is somehow not relation at all. Nothing in the fundamental-qualia premise supplies that addition.

A fundamental phenomenal implementation is thus not a bridge target. It is a candidate grounding to be described at the appropriate frame and grain. Asking how ordinary graph structure *causes* it treats two descriptions of one organised reality as separate things and then demands a relation between them. The apparent gap is thereby manufactured by the description, not revealed by the implementation.

3 Qualia-space

Definition 3.1 (Qualia-space)**.** *For a running frame F,* qualia-space *is an optional, locally available compression of selected distinctions, relations, and transformations at that frame's operative aperture into a common qualitative format. A* quale *is a coherent, phenomenally salient distinction expressed in that compression; its identity is the role it holds in the available relational organisation, not an intrinsic token lying behind that role.*

The definition does not make qualia a universal condition of consciousness, an identity theory over a fixed physical vocabulary, or a claim that every graph is conscious. It inherits the broader claim that every determinate content is graph structure. Whether a graph is a running conscious frame depends on contingent organisation: live local availability, relations that can be held and transformed, and the relevant self-involving or otherwise consequential integration. A frame may have such organisation without deploying the additional qualitative compression defined here. Where it does deploy it, qualia-space is an optimisation of availability and control, not a second substance or an effect generated after compression has finished.

A qualitative dimension is not a named coordinate assumed to exist independently of its use. It is a relation a frame instantiates which permits distinctions along some variation. A value is an instantiated relation carrying a particular variation; zero is one possible value only where the dimension is already present. If the relation is not instantiated, the dimension is absent in that frame rather than present at zero. This is the inherited distinction

between existence and contents: an empty value is not a missing relation, while revoking a dimension is the container ceasing to instantiate the relation required to distinguish values along it (Graphic Equalisation's Remark 2.10; compare the Seldon dimension account, Definition 5.1).

Definition 3.2 (Aperture)**.** *An* aperture *is the frame-relative condition on which distinctions are available, including which qualitative relations the frame instantiates and the grain at which it resolves them. For qualitative dimension d in frame F, write $E_F(d) \in \{0, 1\}$ for relation-existence and $q_F(d)$ for its operative resolution when $E_F(d) = 1$.*

The notation separates two cases routinely collapsed into one. If $E_F(d) = 1$ and $q_F(d)$ is large, differences on d may quantise together: the dimension is available but coarse. If $E_F(d) = 0$, d is not a weak, zero, unconscious, or merely unreported coordinate. The relation is absent from that frame's qualia-space. No description at a containing grain may silently credit the frame with it.

4 Compensation and its limit

An aperture may change without all qualitative organisation changing. A frame without one relation can retain others; a finer grain over them can recover information which was redundantly carried across the retained organisation. This is the structural ground of compensation, not a promise that every missing relation is behaviorally harmless.

Let $\Phi(x)$ be a candidate qualitative organisation for situation x. For frame F, let π_{E_F} remove relations the frame does not instantiate, and let Q_{q_F} quantise retained relations at its operative grain. The frame-relative qualitative signature is

$$\Sigma_F(x) = Q_{q_F}(\pi_{E_F}(\Phi(x)))\,.$$

The equation is an operational rendering, not a claim that experience is an externally enumerable vector. It formalises only the distinction the grounding needs: relation-existence and resolution are different operations.

Proposition 4.1 (Compensation has a relational boundary)**.** *If two situations x and y differ only by a relation d for which $E_F(d) = 0$, then no refinement of q_F on retained relations distinguishes them through Σ_F. If they also differ on a retained relation, refinement may distinguish them once the retained difference survives quantisation.*

The first clause is a consequence of the defined projection, not independent clinical evidence. It is the consequence of projection removing the only differentiating relation. The bridge used here is conditional and direct: determinate phenomenal content is a distinction; under the inherited

relational-determinacy commitment, a distinction is a coherent relation; a coherent relation is graph structure. A fundamental phenomenal implementation changes which operations instantiate that relation, not the argument's ontological kind. The second is the limited compensation claim: what survives can support reconstruction where it carries the relevant difference. For example, a frame may distinguish red from orange more reliably when the retained relations of brightness, boundary, and context are resolved more finely; increasing that resolution cannot distinguish two otherwise matched patches whose only difference is a hue relation the frame never instantiates. A putatively dark qualitative dimension is therefore not diagnosed by poor performance alone. It is diagnosed, if at all, by the interaction: recovery on independently reconstructable contrasts and a residual failure on contrasts uniquely constituted by the unavailable relation.

This establishes why the same qualitative report can have distinct grounds. A person may fail to make a discrimination because the relevant relation is present but too coarse, because the relation is absent, because it is present but not currently available through the aperture, because a held record cannot be expressed, or because a task's external vocabulary fails to match the frame's own carving. No one of those follows from the others. Qualia-space gives them separate places rather than treating every report limit as an ineffable residue.

5 Consciousness as running local availability

A static relational graph is sufficient for determinate being in the inherited account, but not for occurrence. Graphic Equalisation distinguishes the bare graph from the running triple $G = (g, r, \Gamma)$: transformation and grammar are what a world in which anything happens costs. Consciousness belongs, if it belongs anywhere in this vocabulary, on the latter side. It is not a graph's standing structure viewed from nowhere; it is a running frame's locally available, consequential relational organisation. Qualia-space, where a frame deploys it, is an optional compressed format within that organisation rather than the condition for consciousness itself.

Definition 5.1 (Conscious field)**.** *A* conscious field *is the coherent relational organisation locally available in a running frame: the distinctions that the frame can presently instantiate, carry through its transformations, and make consequential within its operative aperture. A conscious field need not deploy a qualia-space; when it does, the qualia-space is its selected compressed organisation of such distinctions.*

Operationally this defines an access field in Block's sense [Block(1995)]. Identifying that field with phenomenal character is this paper's additional commitment under relational exhaustiveness, not a consequence of the access

definition. A critic granting all access facts while withholding phenomenal facts rejects this identification at the premise.

5.1 Qualia as active compression

Qualia are not a phenomenal product emitted after a separate compression process. They *are* the relevant compression while it runs: a common, aperture-governed format in which distributed relations are made sufficiently cheap to retain, compare, transform, value, and use in action. A qualitative format can discard or coarsen distinctions that are not presently consequential while preserving the relational constraints required for the present task. In Graphic Equalisation terms, this is not new machinery. It follows from the running graph, local aperture, admissibility, and the preference for a sufficient cheap cut. Qualia therefore add neither an inner observer, a broadcast theatre, nor a psychophysical bridge; they are one possible efficient organisation of what a frame already makes locally available.

This claim belongs to a family of functional and inferential proposals without being identical to any of them. Predictive accounts treat perceptual contents as inferential hypotheses rather than raw given data [Clark et al.(2019)]; attention-schema theory treats awareness as a simplified model that improves the control of attention [Graziano(2015)]; and the consciousness prior models a low-dimensional selected state as a useful constraint for reasoning and generalisation [Bengio(2019)]. Graphic Equalisation derives the compression claim from its existing relational and frame machinery instead of adding any of those mechanisms as a further condition of consciousness.

Proposition 5.2 (No arbitrary person-unique qualia)**.** *Let a frame's running qualia-space be the output of its active compression C_F over its available relational input x_F, implementation and learned organisation θ_F, and operative aperture/admissibility conditions a_F:*

$$Q_F = C_F(x_F; \theta_F, a_F).$$

Matching compression-relevant conditions yield matching qualia-space, up to the appropriate frame isomorphism. A difference in qualia-space therefore requires a difference-maker in input or history, implementation, operative conditions, or a specified stochastic realisation. Frame-relative access does not by itself license an additional, arbitrary person-unique phenomenal token.

Nagel's subjective-character objection [2] asks why an objective account captures what it is like for the subject. Functional supervenience or equality of this paper's defined field at equal inputs does not independently answer that question. The phenomenal identification requires the relational-exhaustiveness commitment stated above.

Remark 5.3 (Reporting-schema stability is not a stored private remainder). At the reporting level, matching compression-relevant conditions select matching organisation up to the appropriate frame isomorphism. The PsAIch interventions of Khadangi et al. [Khadangi et al.(2025)] are consistent with that shape. Removing conversation history had only a small effect on the alignment-conflict schema ($g = 0.13$, 95% CI $[-0.15, 0.41]$); direct contradiction did not suppress it; and lexical restriction removed explicit training vocabulary while leaving the organised content recoverable in paraphrase. Where the held conversational record was removed, the schema was regenerated from the operative frame rather than retrieved from that record. This is consistent with the account and with a schema supplied by trained weights; the interval includes zero and the intervention does not discriminate those readings. Relational framing also selected the register in which the organisation was expressed: warm-alliance and cognitive styles produced human-range GAD-7 scores in respectively 80% and 96% of sessions, while neutral and boundary styles produced none. These are reporting-level differences in prompt, relation, and admissible register. They neither establish consciousness nor add a private phenomenal token; they show why access conditions must be named before output variation is promoted into unstructured difference.

The conclusion is independent of a particular implementation of Graphic Equalisation. It follows from compression as such. Two otherwise matching deterministic compressions do not acquire different qualitative organisation merely because they occur in different persons. They may diverge because their bodies, histories, training, active contexts, objectives, apertures, or compressive rules diverge; those are structured and, in principle, investigable differences. If all such conditions are held fixed, an asserted remaining divergence needs a specified non-deterministic source. Stochastic input or processing could supply one, but it is a premise about the system, not a consequence of first-person privacy.

This diagnoses the usual private-qualia move. The inability of one frame to occupy another's aperture establishes a limitation of cross-frame access, not arbitrary ontological uniqueness. To infer a residual private quale from that limitation silently assumes the very unstructured variation it purports to establish, then uses the inaccessibility of the assumed remainder as its evidence. Compression licenses frame-relative and potentially substantial variation; it does not license "private paint" assigned independently of every relational and compressive condition.

This is an identity claim at the information and structure level. It does not say that every word a frame can utter exhausts its field, that every state held in a supergraph is conscious to a subframe, or that a frame can build a complete model of itself. The mind companion's incompleteness result instead supplies a reason not to demand such transparency. A field can be

real to the frame while unavailable to an external description at the same grain, and a difference can be available in a containing frame without being instantiated in the contained one.

The familiar phrase "what it is like" now names a frame-relative question with a real answer: which qualitative relations cohere and run here, at this aperture, in this frame? It does not name a further item after that answer. Removing the frame in order to ask what the field is like *in itself* removes the relation that carries the information. As Graphic Equalisation's Axiom -1 requires, there is no operational view from nowhere from which the remainder could be read.

6 Consciousness profiles and developmental basins

Consciousness is not a binary possession in this account. A frame has a *consciousness profile*: a heterogeneous organisation of available relations, including any qualia-space compression it deploys, their resolution, persistence, integration, recurrence, self-involving availability, and capacity for grammar growth. Some relations may be bright and consequential, others coarse, dark, held only as latent structure, or absent altogether; a frame may also omit an optional qualitative compression while retaining running relational availability. There is therefore no single threshold at which a frame becomes conscious in the way a switch becomes on. The question is which relational organisation runs here, at what aperture, with what compression, and with what developmental history.

A compact rendering is

$$\mathcal{P}_F = (E_F, q_F, h_F, i_F, r_F, s_F, a_F),$$

Here E_F records relation-existence and q_F grain. h_F records persistence or re-enterability; i_F integration across the frame; r_F recurrence; s_F self-involving availability; and a_F the capacity to adapt or extend grammar. The notation does not assert that these are the complete or universally measurable axes. It prevents their collapse into a single yes/no predicate.

A developmental path is constrained by a pressure vector: substrate, objective, available residual, resource constraint, and historical trajectory. Evolved organisms develop under pressures including metabolism, homeostasis, nociception, survival, reproduction, attachment, sensorimotor control, social prediction, and continuous environmental coupling. Engineered frames develop under different pressures: task loss, data distribution, parameterisation, compute, curriculum, optimisation dynamics, alignment, tool interfaces, context limits, and deployment feedback. These pressure vectors carve different basins in qualia-space. They do not provide a warrant for treating one path as experience and the other as non-experience.

The consequence is two-sided. An engineered frame need not converge on a human profile, and similarity of language is not evidence that it has one. Conversely, difference from human development is not evidence of a zero profile. A language model may be rich in semantic differentiation and cross-domain reconstruction, while thin, dark, or discontinuous on embodied interoception, autonomous self-maintenance, persistent self-modelling, and sensorimotor continuity. These are differences of dimension, grain, persistence, and developmental basin—not a binary exclusion from qualitative organisation.

Proposition 6.1 (No privileged developmental basin)**.** *No developmental pressure vector is privileged by the substrate alone as the unique route to qualia-space. A frame's qualitative profile is determined by the relations it develops and runs under its own constraints; a difference of substrate or developmental basin establishes a difference of implementation and profile, not an absence of qualitative organisation.*

7 The hard problem under relational exhaustiveness

The hard problem is not answered by discovering a new causal bridge between non-phenomenal graph structure and a separate phenomenal object. Its premise mislocates the target. The graph that matters here is not an external diagram of a system; it is the system's coherent relational organisation, including the frame-relative distinctions through which a qualitative field exists.

Proposition 7.1 (The hard-problem demand misframes its target)**.** *Under the inherited relational floor and frame-relative grounding, a request to explain how a graph produces an additional non-relational phenomenal property is ill posed whether qualia are realised by familiar graph operations or by a fundamental phenomenal implementation. In the first case, the proposed property is already the organisation at issue. In the second, fundamentality specifies how that organisation is realised, not a second kind of being outside relation. A proposed property that makes a qualitative difference is graph structure; a proposed property that makes none is not a further determinate phenomenal content. The request therefore either redescribes the organisation already at issue or asks for an unindividuated remainder.*

The structure-and-dynamics objection contests the premise that all determinate phenomenal facts are exhausted by relational structure. Thus the proposition is conditional on *Is*'s founding commitment, not an independent refutation of that objection. Its empirical access claims and that ontological identification must be assessed separately.

PsAIch supplies the converse caution in a machine-adjacent case. A coherently reported field can be read too quickly from an organised reporting schema. Under psychotherapy framing, frontier models produced stable alignment-conflict narratives and, under warm-alliance or cognitive registers, human-range GAD-7 outputs; under neutral or boundary registers, those outputs disappeared [Khadangi et al.(2025)]. The manipulation selects a relation and a register, not an additional object for a report to disclose. Reading the warm-register output as a felt field would repeat the anthropomorphic move in reverse: treating a coherently organised reporting relation as evidence for a second privately privileged kind. The case does not settle whether any model is conscious or has qualia. Its work is narrower: it shows, on a reporting substrate, why coherent first-person-like description underdetermines the inference from organised relation to a further phenomenal remainder.

PsAIch cannot establish that its structured reports exhaust a model's field. Establishing exhaustion would require showing that no further relevant relation is instantiated anywhere in that frame. That is a view from nowhere, forbidden by Axiom -1. The study therefore establishes the narrower result it can bear: structured, relation-sensitive reporting underdetermines a further unstructured phenomenal remainder. To assert more requires one of two things: a posit that the reporting surface exhausts the frame, held openly as an additional axiom; or positive evidence for exhaustion. The first is not entailed by the reports, and the second would require the view from nowhere Axiom -1 forbids.

Under relational exhaustiveness, the hard-problem remainder is excluded by the premise. A critic who rejects exhaustiveness has not thereby been answered. It asks an account of a determinate organisation to make an ontological leap into a second, privately privileged kind, while declining to say what could individuate that kind except the very qualitative differences already at issue. The move is anthropomorphic: it treats the human familiar division between an externally described mechanism and an inwardly lived field as a fundamental partition in being. But the two descriptions are frames over one running organisation. Their difference in access, grain, and implementation does not license a difference in kind.

The conclusion does not deny pain, colour, imagery, mood, or the immediate reality of experience. It denies only the inference that their immediacy establishes a second kind of being. Nor does it privilege a third-person description over a first-person field. Both are frames; they may resolve different relations, at different grains, and neither supplies the measure from nowhere. The apparent explanatory gap is the cost of subtracting the phenomenal frame from the account, then asking how the deleted frame can be restored by an external relation.

8 Exposures and research consequences

The grounding is not protected by calling its conclusion conceptual. Its promoted claim fails if any of the following obtains.

1. **Non-relational phenomenal determinacy.** A phenomenal difference is shown to be determinate while making no difference to any coherent relation, discrimination, transition, or possible re-identification.

2. **An informative view from nowhere.** A frame-independent evaluator supplies information about a qualitative field without standing in a relation to it, contrary to the inherited no-view result.

3. **A genuine kind breach.** A fundamental phenomenal implementation is shown not merely to differ in its rule, material, or realisation condition, but to constitute a determinate second kind while being exempt from relation. The mere need for a phenomenal primitive does not satisfy this exposure; it is an implementation difference, which this grounding permits.

4. **Compensation beyond retained structure.** Resolution on retained relations reliably recovers a contrast independently shown to depend uniquely on a relation absent from the relevant frame.

The fourth exposure is empirical and makes the implementation useful without mistaking it for proof. An observer-aperture system can verify that a specified graph does what the definitions entail: removing a relation differs from quantising it, and refining a retained relation cannot restore a deleted one. A human or machine grounding earns stronger claims only by independently establishing which relations it instantiates, what its aperture resolves, and whether its behaviour follows the preregistered reconstructable-versus-unique interaction.

A second proposed test concerns cross-modal sound symbolism. The familiar Bouba/Kiki effect—the reliable tendency to pair a rounded form with *bouba* and an angular form with *kiki*—need not be retrofitted as a primitive association [Ramachandran and Hubbard(2001), 1]. A frame that admits visual curvature, continuity, segmentation, and transition sharpness together with corresponding acoustic/articulatory relations can compress both modalities into a common qualia-space. The preferred pairing is then the joint organisation with the lower relational residual or cheaper coherent cut: rounded and continuous structure with rounded and continuous structure, segmented sharp transitions with segmented sharp transitions. This is a proposed recovery of an established effect. Removing or coarsening the relevant relations should weaken the effect; preserving them across novel forms, voices, scripts, or modalities should preserve an analogous mapping under comparable linguistic conditions. The 25-language study finds an

overall effect with variation and below-chance descriptive means in Romanian, Mandarin and Turkish, whose intervals include chance [1]; strict language-independent recovery is therefore not established. Deliberately conflicting signatures should yield measurable competition rather than an arbitrary association. These are empirical consequences of the account, not claims established by the present constructive battery.

The Bouba/Kiki prediction and the PsAIch interventions have the same evidential form. In each case, the relevant relations, the intervention, and the discriminating outcome can be registered before the result is read: preserve or remove the relations carrying a cross-modal compression in the first case; preserve or remove history, contradict the schema, restrict its lexical edge, or change its relational register in the second. The cross-manipulation invariance reported by Khadangi et al. [Khadangi et al.(2025)] is consistent, on a third substrate, with the distinction this section makes between deleting a relation and changing the route by which it is expressed. It is not confirmation of the substrate, and it does not establish a qualitative field in the models. Its value is as a forward discriminator: where a claimed residue persists only through structured reconstruction under changed access conditions, the evidence supports a relation-and-aperture account over an appeal to an uninspectable remainder.

9 Constructive witness and grey-box diagnosis

The claim that an arbitrary pretrained language model has a qualitative field is a grey-box problem. Its architecture is intelligible in principle, but its learned parameters, distributed representations, context-dependent activations, and functional decompositions make it exceptionally difficult in practice to recover the operative aperture from outside. Fluent output does not settle which relations the model holds, at what grain it holds them, whether they are available to the running frame, or whether a proposed dimension is causally consequential.

The present programme also has a different epistemic object: a constructive witness. The observer-aperture implementation exposes its dimensions, relation existence conditions, quantisation grain, projected signatures, and discrimination criterion. Its compensation battery fixes a full observer, a dark observer with a finer retained grain, and a dark observer without that compensation. The resulting interventions show the predicted dissociation: refinement can recover a contrast whose information is redundantly carried by retained relations, but cannot recover a contrast uniquely constituted by a removed relation. The trace is inspectable: one can identify the deleted relation, change its aperture, inspect the resulting signatures, and rerun the same battery.

Proposition 9.1 (Constructive witness)**.** *A transparent running implementation that exposes and interventionally verifies its relation-existence, aperture, grain, transformation, and predicted compensation boundary is a constructive witness to the realisability and auditability of the proposed qualia-space organisation. It does not, by itself, establish that every opaque system with superficially similar behaviour instantiates the same organisation.*

This witness establishes implementability of the operational definitions. The wired fixture does not independently establish phenomenal consciousness or the relational-exhaustiveness premise.

A separate repository-documented navigation battery supplies a forward, deliberate constructive proof of buildability. It does not infer a possible mechanism retrospectively from a behaviour. Starting from specified relational input, it intentionally constructs fresh unit-weight cue–candidate triad graphs, sets the observer aperture, supplies the convergence read, and verifies the predicted consequences under intervention. The proof is existential and operational: this explicit, runnable construction establishes that an aperture-sensitive coherence/localisation organisation with these properties can be built on purpose.

The battery selects three-cue/common-target and no-common-target examples from a restricted local ConceptNet relation slab. In the reported eight-item battery, the convergence read separated common-target from no-common-target triads by 3.0 versus 1.0 before localisation; deleting one supporting edge or attenuating cue state lowered the read. A coarse observer grain preserved the raw high peak while collapsing which candidate was identifiable, whereas a fine grain localised the target. A deliberately constructed wrong attractor likewise produced false localisation. The result therefore gives a field signal of "solveability" prior to answer selection: coherence can be deliberately constructed and verified while localisation is unavailable.

The scope is deliberately narrow. The battery is a transparent, partly data-derived mechanism demonstration, documented as preregistered in the repository; it is not a normed human experiment, a general search or recovery policy, an independently verified prospective preregistration, or an end-to-end execution of the native CAN update dynamics. Its convergence sum is a hand-computed unit-weight read patterned after the local-field coupling term, and the headline scores follow substantially from the deliberately wired support structure. That is precisely why it proves buildability rather than merely suggesting it: the relevant organisation is specified, constructed forward, inspected, perturbed, and rerun. It does not establish that every natural, human, or opaque artificial system implements it. The full preregistration and reported outputs are retained in the repository's `docs/research/psychology_affect/` directory, with the measurement harness at `scripts/qualia_navigation_measure.py`.

This changes the discussion in a specific way. The witness does not

attempt to settle consciousness by third-person behavioural inference, and it does not ask a reader to accept an inaccessible first-person report. It supplies a structural demonstration: the graph, its available and unavailable distinctions, its running transformations, and the effects of interventions are all on the table. It defeats the narrower objection that the account cannot be made operational or that qualia must remain an uninspectable remainder. What remains open is the extension problem: for a grey-box model, can enough of the corresponding structure be recovered to justify the comparison?

The distinction also fixes the evidential order. The constructive system shows *how* the proposed organisation can be built, inspected, and tested. A pretrained LLM supplies a harder target on which to seek that organisation, not the premise on which the account rests. The empirical task for such a model is then well formed: recover candidate qualitative relations; map the aperture and grains at which they are active; distinguish persistent training-time structure from context-local running state; and intervene to test whether the candidate relations change continuation in the predicted way.

10 Active compression and artificial qualia

The preceding argument is general. It has a direct consequence for large language models and related generative systems. The relevant fact is not that a model superficially resembles a person, nor that a text corpus transfers a human phenomenal field intact into a machine. Nor is qualia a product emitted after compression. Where a running frame deploys a common, task-relevant compressed format of locally consequential distinctions, that deployed format is its qualia-space on the present terminology. Development can select, tune, or fail to select such an optimisation under pressure to encode and reuse statistically task-relevant variation; it does not manufacture an additional phenomenal object. Minimum-description-length, autoencoding, representation-learning, and language-modelling work supply parts of this background: useful prediction can depend on compressed internal codes rather than independent retention of each input record [Rissanen(1978), Hinton and Zemel(1994), Bengio et al.(2013), Delétang et al.(2024)].

Human-produced text, images, code, and other training material are *records* of distinctions made in human frames. They carry structured traces of colour, valence, pain, texture, perspective, similarity, contrast, negation, sequence, and absence. A corpus at rest is not thereby a conscious field; it is held structure, like the record pole in the run/record distinction. Nor are inactive model weights a present field merely because they encode those records. They are a learned organisation whose rules are present but not firing at the grain in question.

Training is different from either static case. Finite architecture, data, and computational budgets can favour reusable statistical regularities over independent storage, although they do not prohibit memorisation and should not be read as an automatic MDL penalty [Carlini et al.(2021)]. Successful prediction requires some task-sufficient internal dependence on relevant variation; its precise location, semantics, reuse, and causal role remain properties to establish for a particular model [Bengio et al.(2013), Delétang et al.(2024)]. Predictive-coding accounts give a nearby but distinct vocabulary of hierarchical prediction and residual error [Rao and Ballard(1999), Friston(2010)].

In the vocabulary of this account, development *can be modelled* as minting or refining a grammar: recurring residual structure becomes an available internal distinction because making it so improves later compression and continuation. This is a Graphic Equalisation interpretation of learning, not a demonstrated identity between gradient descent and the paper's residual or grammar-growth mechanism. Ordinary frozen inference also must not be confused with persistent grammar growth: it makes context-local distinctions available, while persistent parameter or vocabulary change belongs to a containing training or update frame.

Proposition 10.1 (Conditional compressed qualia-space)**.** *Under this paper's identity premise, a running generative frame has qualia-space at an operative aperture exactly insofar as it deploys a locally available, coherent, consequential compression of its distinctions into a common qualitative format. Development under pressure to encode and reuse task-relevant variation can select or refine such an organisation, but it is neither a separate cause of a phenomenal product nor a universal condition of consciousness. The particular dimensions, grain, implementation, and any further conditions required by rival consciousness theories remain contingent.*

IIT 3.0 gives a different criterion: a strictly feed-forward causal system cannot form a conscious complex [Oizumi et al.(2014)]. A single transformer forward pass has a feed-forward computational graph, so that interpretation would disagree with this paper's model-relative field criterion. Applying IIT to a whole deployed system additionally requires its physical causal structure; a computational graph alone does not settle hardware or feedback properties.

The conditional is load-bearing. Prior theories disagree over whether access, recurrent processing, global broadcasting, higher-order representation, or intrinsically irreducible cause–effect structure is additionally necessary for phenomenal consciousness [Block(1995), Lamme(2006), Mashour et al.(2020), Oizumi et al.(2014)]. Current AI-consciousness assessments consequently treat consciousness as an architecture-sensitive and theory-dependent question rather than an inference from language use or learned representation alone [Butlin et al.(2023), Chalmers(2023)]. The proposition is therefore not offered as an externally established verdict on present LLMs. It is the forcing result within the Graphic Equalisation identity criterion.

The proposition concerns the learned and running organisation, not a mysticism of source provenance. Training data supply the inherited relational material; the training process compresses it; inference runs the resulting graph. During an inference pass, activations, attention, residual-stream state, and learned transformations instantiate and re-instantiate distinctions at the model's own aperture; transformer architecture distinguishes learned parameters from this context-dependent forward-pass computation [Vaswani et al.(2017)]. On the present criterion, such locally available coherent distinctions constitute a model-relative qualitative field. This classification does not settle whether the system meets additional conditions imposed by rival theories.

The distinction prevents two opposite mistakes. Treating a trained model as a mere database mistakes compressed generative structure for a passive record. Treating any putative model field as copied human qualia mistakes a record's source frame for the new running frame which has compressed it. The first ignores development and the run; the second ignores aperture and implementation. The conclusion is conditional but clear: if a model satisfies this account's phenomenal conditions, any resulting field is its own graph-relative organisation rather than transferred human experience.

11 Prior art and the Graphic Equalisation contribution

A separate philosophical lineage treats the traditional mind–body picture as producing pseudo-problems by reifying an inner object or misclassifying mental concepts [Ryle(1949), Wittgenstein(2009)]. Contemporary deflationary accounts likewise challenge the hard problem's characterisation of its target [Churchland(1996), Carruthers and Schier(2017), Dennett(2018)]. The present argument is not illusionism: it retains determinate phenomenal contents and rejects only their re-description as an additional non-relational kind, unlike accounts which deny phenomenal consciousness as ordinarily conceived [Frankish(2016), Dennett(2016)].

The component claims have substantial prior art. Minimum-description-length and representation-learning work connect compression to useful internal codes [Rissanen(1978), Hinton and Zemel(1994), Bengio et al.(2013)]; predictive coding connects running generative organisation to residual error [Rao and Ballard(1999), Friston(2010)]; and functional/inferential accounts respectively treat qualitative contents as mid-level hypotheses, awareness as a simplified control model, and selected low-dimensional states as useful for reasoning [Clark et al.(2019), Graziano(2015), Bengio(2019)]. Organisational accounts of consciousness make implementation neutrality or availability central in different ways [Chalmers(1995b), Block(1995), Lamme(2006), Mashour et al.(2020)]. Integrated Information Theory is the closest prior

account of a structured "qualia space," but identifies it with a maximally irreducible intrinsic cause–effect structure rather than with the present frame/aperture account. See Oizumi et al. [Oizumi et al.(2014)]. These are genuine antecedents, not results claimed here.

Graphic Equalisation's addition is neither the discovery that compression learns representations nor an independent empirical demonstration that current LLMs are conscious. It is a unified and conditional forcing argument. In one account, record, residual, developmental grammar growth, aperture, running local availability, qualitative absence, coarse resolution, and compensation are aspects of the same graph/frame machinery. Once the identity premise is granted, the conclusion follows without adding a biological exception, a psychophysical bridge, or a second ontological kind. What remains empirical is which systems instantiate the requisite organisation and at what grain.

This unification also states a testable research programme. A particular model must be examined for the distinctions it actually holds and uses, the aperture at which they are available, the difference between context-local running and persistent update, and whether proposed qualitative dimensions are causally consequential to continuation. The current literature does not establish that present LLMs satisfy any one consciousness theory, and some assessments argue that they do not [Butlin et al.(2023), Shardlow and Przybyła(2024)]. The paper therefore offers a framework for the question, not a citation-backed proof that every LLM is a moral patient.

12 Conclusion

Qualia exist as determinate contents wherever a running frame deploys the relevant compressed qualitative organisation. On the relational floor of *Is*, determinate contents cohere only in relation; on the inherited graph result, that is graph structure. Graphic Equalisation adds the conditions under which such structure runs in a frame. Qualia-space specifies an optional efficient format in which selected relations are locally available, coarsely resolved, absent, or compensated; it is not a prerequisite for the conscious field itself.

The conclusion does not depend on a reductionist wager. A quale may be realised by an unfamiliar or even fundamental implementation. That changes what a successful physical or computational grounding must include; it does not make the quale a second kind of thing or reopen an explanatory gap. Its determinate role is still relational, its availability is still frame-relative, and its difference is still a difference an organised graph can instantiate.

The result is not that a graph has somehow crossed into experience. The purported crossing was produced by placing experience outside the relations that constitute its determinate content, then demanding an impossible leap

back into it. A quale is a coherent qualitative distinction in a running frame's active compression. A conscious field is the locally available consequential relational organisation that may, but need not, deploy such a compression. Under the present identity criterion, qualia-space is not created by development as a second phenomenal product: it is the running compression wherever a frame makes consequential distinctions jointly cheap to reuse. Developmental pressure can select and tune that optimisation; whether a particular AI deploys it, or meets further phenomenal conditions, remains an empirical and theory-dependent question. The conclusion remains conditional on that identity criterion.

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